

MarketMancer: A Multi-Layer System for Factor Construction, Forecasting, and Alpha- Based Portfolio Optimization

Team 20

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Abstract

This project develops MarketMancer, a unified decision support system that links macro conditions, factor level signals, stock specific characteristics, and long / short portfolio construction in a single framework. The goal is to give investors one engine that can move from regime aware factor timing down to stock selection and portfolio weights, instead of treating those steps as separate workflows.

The forecasting layer was initially built around a macro informed machine learning ensemble. Factor premia were modeled using Ridge, Lasso, and Random Forests on a rich set of state variables, including yield curve shape, credit spreads, volatility indices, and lagged factor performance. Only later, as a robustness check, did we add a comparatively simple AR(1) benchmark. Walk forward validation showed that this benchmark systematically outperformed the ensemble on directional hit rate and forecast error, across all four factors and all three testing regimes. As a result, AR(1) now serves as the primary factor premia forecast, while the macro ensemble is kept as a diagnostic layer that explains how macro variables load onto each factor.

On top of these forecasts, MarketMancer estimates stock level expected returns with a pooled Lasso model on standardized fundamental and technical features, measures risk using a Ledoit-Wolf shrinkage covariance estimator, and constructs a constrained long / short portfolio that respects position caps and gross exposure limits. Across the small, medium, and large regimes, the system produces economically intuitive factor behavior, stable and interpretable forecasts, and unified portfolios with positive expected H -day alpha, strongest in the largest sample. The main lesson is not that “simple is always better,” but that genuinely testing sophisticated models against disciplined baselines can reveal when added complexity improves a forecasting process and when it does not, all within a transparent, "glass box" architecture.

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1. Introduction

1.1 Background and Motivation

The quantitative investment industry has undergone a remarkable transformation over the past four decades, evolving from simple rule-based trading systems to sophisticated multi-factor models that integrate diverse data sources, advanced statistical techniques, and machine learning algorithms. This evolution has been driven by several concurrent developments: the exponential growth in available financial data, dramatic improvements in computational power, advances in statistical and machine learning methodology, and the democratization of financial APIs that provide institutional-quality data to a broader audience.

Factor investing, which forms the theoretical foundation of this project, emerged from the empirical observation that certain characteristics of securities, such as their valuation ratios, profitability metrics, recent price momentum, and volatility profiles, systematically predict future returns. The intellectual origins of factor investing can be traced to the Capital Asset Pricing Model (a.k.a. CAPM) developed by Sharpe (1964), Lintner (1965), and Mossin (1966), which proposed that a single factor, market beta, explains the cross-section of expected returns. However, subsequent empirical work revealed that CAPM's predictions were inconsistent with observed return patterns.

Fama and French (1993) extended this framework with their seminal three-factor model, demonstrating that size (SMB, or Small Minus Big) and value (HML, or High Minus Low book-to-market ratio) factors explained substantial variation in equity returns beyond market beta alone. This work fundamentally changed how practitioners and academics understood equity risk and return. The three-factor model

was later extended to include momentum (Carhart, 1997), profitability and investment (Fama and French, 2015), and numerous other factors that have been proposed in the academic literature.

The rapid growth of published return predictors, often referred to as the “factor zoo” (Cochrane, 2011), has created both opportunities and challenges for investors.

Academic research now contains hundreds of signals that appear to forecast returns, but this large menu also raises concerns about data mining and the reliability of many reported results. Harvey, Liu, and Zhu (2016) argue that a significant share of published factors may be statistical false positives, and they recommend much stronger statistical standards before accepting a new factor as genuine. McLean and Pontiff (2016) offer a complementary perspective. They show that anomaly returns tend to decline after the underlying research becomes public, which is consistent with investors learning about and trading on the signals. However, they also find that these strategies continue to generate returns that are positive and economically meaningful. Together, these findings suggest that the factor space is promising, but careful validation and skepticism are essential.

Against this backdrop, modern quantitative investment systems must navigate several practical challenges. First, they must select from the universe of documented factors those that are most likely to persist out of sample and that are implementable given transaction costs and capacity constraints. Second, they must integrate factor views at the portfolio level while accounting for factor exposures at the individual security level, a challenge that requires bridging the gap between “top-down” macro factor timing and “bottom-up” stock selection. Third, they must make forecasts under uncertainty while acknowledging that financial market dynamics are non-stationary and that historical relationships may not persist.

The system presented in this capstone project, MarketMancer, is designed to explore these questions while addressing a broader structural gap in how investors typically interact with quantitative information. MarketMancer brings together four well-established equity styles (Value, Quality, Momentum, and Low Volatility), conditions factor premia forecasts on observable macroeconomic state variables, tests both machine-learning-based and statistical forecasting approaches, and generates stock-level alpha estimates that flow directly into a unified long-short portfolio. In addition to unifying macro, factor, and stock-level perspectives, the project provides a natural testbed for evaluating whether macro-aware ML models can meaningfully improve short-horizon factor timing, and how such models compare to disciplined statistical baselines when evaluated in a rigorous walk-forward framework.

The motivation for this project arises from a common frustration among investors: the difficulty of finding a single tool that simultaneously considers macroeconomic conditions, factor-level views, and stock-specific characteristics. Most existing systems operate at only one level of this hierarchy. Macro strategists focus on top-down regime identification but may lack the granularity for stock selection. Factor investors build portfolios based on characteristics but often treat factor premia as static rather than time-varying. Stock pickers focus on individual company analysis but may fail to account for systematic factor exposures. MarketMancer integrates all three levels, providing a coherent framework that connects macroeconomic conditions to factor attractiveness and ultimately to specific portfolio recommendations.

The system is implemented as a web-based application using Streamlit, with a modular Python backend that handles data collection, factor construction, forecasting, and optimization. Data is sourced from Financial Modeling Prep (FMP), a commercial API that provides fundamental financial statement data, price histories, and macroeconomic indicators. The system is designed for transparency and

interpretability, with each component producing outputs that can be examined and validated by the user. This design philosophy reflects a broader conviction that quantitative investment systems should be “glass boxes” rather than “black boxes” and their predictions should be understandable in terms of economic intuition, not merely accepted as outputs of an opaque algorithm.

1.2 Literature Review

This section reviews the academic and practitioner literature that informs the design of MarketMancer, organized around four themes: factor construction and selection, factor timing and regime dependence, stock-level alpha prediction, and portfolio optimization under estimation error.

1.2.1 Factor Construction and Selection

The four factors employed in MarketMancer (Value, Quality, Momentum, and Low Volatility) are among the most extensively documented in the academic literature. Each has a substantial body of evidence supporting its out-of-sample predictive power, although the economic mechanisms underlying each factor differ.

Value Factor. The value premium refers to the empirical pattern that stocks with low prices relative to fundamentals tend to earn higher average returns than stocks with high prices relative to fundamentals. Early evidence related to this idea appears in the work of Sanjoy Basu (1977), who showed that low price-to-earnings stocks earned higher risk-adjusted returns than high P/E stocks. Subsequent research expanded the definition of value beyond earnings. Rosenberg, Reid, and Lanstein (1985) documented strong return differences using book-to-market ratios. The value premium was formally established as a central factor in asset pricing by Fama and French (1992, 1993), who showed that high book-to-market stocks earn persistently higher average returns and constructed the HML value factor. The economic source of the value premium remains debated. Risk-based explanations argue that value stocks

are fundamentally riskier, particularly in bad economic states (Fama and French, 1996; Zhang, 2005). Behavioral explanations instead emphasize systematic investor biases such as over-extrapolation of past growth and institutional constraints on arbitrage (Lakonishok, Shleifer, and Vishny, 1994; Daniel and Titman, 1997).

MarketMancer constructs the value factor using three valuation ratios: book-to-price (BP), earnings-to-price (EP), and free cash flow-to-price (FCFP). For each date, these ratios are standardized across the cross section of stocks and then averaged to produce a composite value score. In addition, the model computes these scores sector by sector whenever sector data is available, so that a “cheap” stock is defined relative to its peers rather than the entire market. Using multiple valuation ratios is common in the factor investing literature, since it diversifies across different aspects of valuation and can reduce reliance on any single accounting measure.

Quality Factor. The quality premium refers to the tendency of high quality companies with strong profitability, solid balance sheets, and more stable performance to earn higher average returns than low quality firms. Novy-Marx (2013) shows that gross profitability predicts cross sectional returns with a magnitude comparable to traditional value effects. Fama and French (2015) incorporate profitability and investment into their five factor model, and Asness, Frazzini, and Pedersen (2019) propose a broad quality factor that combines profitability, growth, safety, and payout characteristics.

MarketMancer’s quality factor follows this multi dimensional spirit. It combines five metrics: return on equity (ROE), return on assets (ROA), gross margin, free cash flow margin, and inverse leverage (the negative of the debt to equity ratio, so that lower leverage receives higher scores). These metrics are standardized and then averaged into a composite quality score, with scores computed within sectors when possible. This construction lets the model capture several different dimensions of firm quality, rather than relying on any single indicator.

Momentum Factor. The momentum effect is the empirical finding that recent winners tend to keep outperforming and recent losers tend to keep underperforming over intermediate horizons. Jegadeesh and Titman (1993) document this pattern using strategies that buy past winners and sell past losers. Unlike value and quality, momentum is a purely price based factor with no direct link to fundamentals. The standard academic implementation uses 12 month returns excluding the most recent month (the “12 minus 1” momentum strategy), which is designed to capture medium term trends while avoiding very short term reversal effects. Behavioral explanations for momentum emphasize gradual investor underreaction to news (Hong and Stein, 1999) and biases such as confirmation or overconfidence (Daniel, Hirshleifer, and Subrahmanyam, 1998). Risk based explanations are more challenging, in part because momentum factor “crashes” tend to occur during sharp market rebounds and stress episodes (Daniel and Moskowitz, 2016).

MarketMancer implements a short horizon momentum measure using 60 day price returns. These returns are converted into sector relative scores so that the model distinguishes true momentum from simple sector rotation. This 60 day specification is a practical proxy for longer horizon momentum definitions that fits the daily data retrieved from the API and keeps the implementation simple and transparent.

Low Volatility Factor. The low-volatility anomaly captures the surprising tendency for lower-risk stocks, whether measured by total volatility or market beta, to earn returns that are as high as or higher than their riskier counterparts. This pattern runs against the basic intuition of the CAPM, which predicts a positive tradeoff between risk and expected return. Evidence related to this appears in Ang, Hodrick, Xing, and Zhang (2006, 2009), who show that stocks with unusually high idiosyncratic volatility go on to underperform, reinforcing the broader idea that risk and return are not moving together in the way simple theory suggests. Baker, Bradley, and Wurgler (2011) offer an institutional explanation: because many investors are benchmarked

and face limits on using leverage, they tend to chase high-beta stocks, leaving low-beta stocks undervalued. Frazzini and Pedersen (2014) formalize this mechanism in their “betting-against-beta” model and show, across global markets, that a leveraged low-beta portfolio earns a persistent premium. Together, these studies build a consistent picture of why low-volatility stocks may outperform despite carrying less conventional risk.

MarketMancer’s low volatility factor uses 60-day realized volatility (the standard deviation of daily returns), inverted and standardized so that lower volatility receives higher scores.

1.3 Problem Definition

First, let us introduce the “story” behind this tool as intuitively as possible with this realistic question from a hypothetical investor:

“Why is it so hard to find a tool that looks at both the big picture *and* the stock-level at the same time? I want something that knows today’s macro and factor environment, understands stock specific alpha, and then actually tells me which factors to lean into, which names to own or short, and how to put it all together in one coherent portfolio.”

This is exactly the frustration our tool is designed to solve.

The central problem addressed by this capstone project is the integration of multiple levels of quantitative investment analysis, including macroeconomic regime identification, factor-level views, and stock-specific alpha prediction, into a unified and actionable decision-support system.

Specifically, the project addresses the following questions:

1. **Factor Construction:** How should equity factors (Value, Quality, Momentum, Low Volatility) be constructed from fundamental and price data to maximize out-of-sample predictive power while maintaining interpretability?
 2. **Factor Timing:** Can factor premia be forecasted using observable macroeconomic variables? If so, which variables are most informative, and how should forecasting models balance accuracy against overfitting?
 3. **Alpha Prediction:** Can stock-level expected returns be predicted using fundamental and technical characteristics? How should prediction models be regularized to avoid overfitting to historical patterns that may not persist?
 4. **Portfolio Construction:** How should factor views and stock-level alphas be combined into an optimal portfolio, accounting for estimation error in inputs and practical constraints on position sizes and leverage?
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2. MarketMancer: System Overview

MarketMancer is designed around a simple idea: investors make better decisions when they can see the market from multiple perspectives at once. Most tools focus on only one slice of the investment process. They either analyze macro conditions, or evaluate factor exposures, or screen individual stocks, or optimize portfolios.

MarketMancer tries to connect all of these pieces in one place and show how they fit together.

At a high level, the system does five things.

First, it constructs factor signals such as Value, Quality, Momentum, and Low Volatility from raw fundamentals and daily prices. For each date, the tool converts accounting items and price history into standardized factor signals across the chosen

universe. These standardized scores feed into long/short factor portfolios and factor premia forecasts. Separately, the system also maintains sector-relative percentiles for S&P 500 stocks, which are used later to interpret individual stock factor scores.

Second, it evaluates how these factor portfolios have actually behaved over time. Before attempting to forecast anything, the tool shows whether the constructed factors reproduce well known patterns: momentum’s trend persistence, value’s cyclicalities, low volatility’s defensive profile, and so on. This step provides a sanity check that the signals behave in a way that makes economic sense.

Third, it forecasts factor premia over a user selected time horizon. MarketMancer compares a simple autoregressive benchmark (AR(1)) with a complex, macro-aware ensemble of machine learning models (Random Forest, Lasso, and Ridge) that incorporates interest rate levels and slopes, credit spread proxies, VIX based volatility measures, and lagged factor performance. This layer answers the question: which factors are positioned to outperform next, given both their recent history and the current macro environment?

Fourth, it predicts stock level alphas using standardized fundamental and technical features. This component moves from “Which factors look attractive?” to “Which individual names within those factors look best right now?” The model combines fundamental signals, recent price behavior, and regularization to produce intuitive, interpretable forecasts of each stock’s expected H day return, alongside a separate fundamental score that summarizes its value and quality profile.

Finally, MarketMancer builds a unified long/short portfolio that ties everything together. Using stock level expected alphas and a Ledoit–Wolf shrinkage estimate of recent covariance, the optimizer assigns long weights to stocks with strong expected returns and short weights to those with negative expected returns, subject to practical constraints on position size and gross exposure. The result is a coherent portfolio that reflects both factor level views and stock specific insights.

These five components form a single, end to end pipeline. The user selects the universe, date range, and forecast horizon, and the system retrieves the necessary data, constructs factors, generates forecasts, estimates alphas, and produces a tradable portfolio. Each part of the pipeline is visible to the user, reflecting the design principle that quantitative tools should behave like “glass boxes,” not black boxes.

The next section describes how this pipeline is implemented at the data and modeling level: where the inputs come from, how they are aligned, and how factor, forecasting, alpha, and portfolio blocks are built in detail.

3. Methodology: From Data to Portfolio

This section describes how MarketMancer turns raw data and user inputs into factor views, stock-level alpha estimates, and a unified long short portfolio. The pipeline has a simple structure: the user chooses a universe and modeling settings, the system collects and aligns data, constructs factor signals, forecasts factor premia, estimates stock-level alphas, and then builds a tradable portfolio consistent with those signals.

At the code level, the system is implemented as a modular Python codebase with a Streamlit front end. Dedicated modules handle data retrieval, factor construction, forecasting, and optimization, while the Streamlit app provides the interface for selecting inputs and viewing results. This separation keeps the pipeline easy to reason about and easy to modify.

3.1 Universe Construction and User Inputs

The starting point is a set of modeling choices the user makes in the app:

- **Universe size.** A slider lets the user select between 10 and 509 tickers in increments of 5. The universe is always drawn from current S&P 500 constituents.
- **Selection rule.** The behavior depends on whether the user checks the box labeled “Randomize universe selection.”
 - If the checkbox is checked, the system samples N tickers at random, using the user provided random seed so results are reproducible.
 - If the checkbox is unchecked, the system takes the first N tickers in alphabetical order from the S&P 500 list.
- **Start date and sample window.** The user selects the earliest date to include in the analysis. The tool pulls data from that date forward to the most recent trading day.
- **Forecast horizon H .** The user chooses an H -day prediction horizon anywhere from 5 to 63 days. This horizon is applied consistently to factor premia forecasts, alpha modeling, and the unified portfolio’s expected H -day return.

The S&P 500 constituent list is retrieved from the FMP `/stable/sp500-constituent` endpoint. If this call fails, the system falls back to a hardcoded list of 509 S&P tickers stored locally so that the app remains usable. Sector and industry labels from FMP are attached to each ticker and are later used for sector-relative diagnostics and S&P 500 benchmark percentiles.

Once the universe, start date, and horizon are specified, the rest of the pipeline is fully automated.

3.2 Data Foundations and Preprocessing

After the universe is set, MarketMancer builds a shared data backbone that all later components use.

3.2.1 Primary Data Source

All data are sourced from Financial Modeling Prep (FMP) through a custom `FMPDataFetcher` client. This wrapper:

- Handles authentication and API keys
- Sends HTTP requests to the relevant endpoints
- Parses JSON responses into pandas DataFrames
- Applies basic error handling and throttling to respect rate limits

FMP provides four main categories of input:

- **Fundamental data:** quarterly and annual financial statements (balance sheet, income statement, cash flow statement)
- **Price data:** adjusted daily closing prices with full history for each ticker
- **Macroeconomic data:** Treasury yields, VIX, and prices of key bond ETFs used to proxy credit spreads
- **Metadata:** sector and industry classifications, which are used for sector grouping and S&P 500 diagnostics

3.2.2 Fundamental Data

For each stock in the chosen universe, the system collects:

- **Balance sheet items:** total stockholders' equity, total assets, total liabilities, total debt, cash and cash equivalents
- **Income statement items:** revenue, net income, gross profit, operating income, earnings per share
- **Cash flow items:** free cash flow, operating cash flow

- **Share data:** shares outstanding and market capitalization (from price times shares)

From these raw items, standard factor-style metrics are constructed:

- **Valuation ratios:**

- Book-to-price:

$$BP_{i,t} = \frac{\text{Equity}_{i,t}}{\text{MarketCap}_{i,t}}$$

- Earnings-to-price:

$$EP_{i,t} = \frac{\text{NetIncome}_{i,t}}{\text{MarketCap}_{i,t}}$$

- Free cash flow-to-price:

$$FCFP_{i,t} = \frac{\text{FreeCashFlow}_{i,t}}{\text{MarketCap}_{i,t}}$$

- **Profitability and margins:**

- Return on equity:

$$ROE_{i,t} = \frac{\text{NetIncome}_{i,t}}{\text{Equity}_{i,t}}$$

- Return on assets:

$$ROA_{i,t} = \frac{\text{NetIncome}_{i,t}}{\text{TotalAssets}_{i,t}}$$

- Gross margin:

$$GM_{i,t} = \frac{\text{GrossProfit}_{i,t}}{\text{Revenue}_{i,t}}$$

- Free cash flow margin:

$$FCFMargin_{i,t} = \frac{FreeCashFlow_{i,t}}{Revenue_{i,t}}$$

- **Leverage:**

- Debt-to-equity:

$$D/E_{i,t} = \frac{TotalDebt_{i,t}}{Equity_{i,t}}$$

- Inverse leverage:

$$LevInv_{i,t} = -D/E_{i,t}$$

Fundamentals are treated in a point-in-time manner: on each trading date, only financial statement information that would have been available as of that date is used.

3.2.3 Price Data

For each universe ticker, the system pulls a daily series of adjusted close prices over the selected sample window. From these prices it computes:

- Daily returns

$$r_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}$$

- Daily log returns

$$r_{i,t}^{\log} = \ln P_{i,t} - \ln P_{i,t-1}$$

- Rolling cumulative returns for momentum windows (for example 5, 21, and 63 days)
- Rolling realized volatility, measured as the standard deviation of daily returns over a fixed window (for example 60 or 63 days)

Log returns are preferred for constructing cumulative returns over an H -day horizon because they are approximately additive over time. All price and return series are aligned on a common trading-day index for the entire universe.

In addition, the system retrieves price data for SPY, which serves as the market proxy in various diagnostics.

3.2.4 Macroeconomic Features

The factor premia models require a set of macro and financial state variables. These are constructed from FMP time series for:

- **Treasury yields:**

- 3-month, 2-year, 5-year, and 10-year yields
- Level: average of available yields on each date
- 1-month change in level: difference in the level over the prior 21 trading days
- Term spreads:

- $$TS * 10y - 2y = Yield * 10y - Yield_{2y}$$

- $$TS * 10y - 3m = Yield * 10y - Yield_{3m}$$

- **Equity volatility (VIX):**

- VIX closing level
- 20-day moving average of VIX
- 252-day percentile of VIX, which locates each day's VIX in its 1-year trailing distribution

- **Credit spreads (ETF based proxies):**

- Long Treasury ETF: TLT
- High yield credit ETF: HYG

- Investment grade credit ETF: LQD
- High yield spread proxy:

$$\text{HY_spread}_t = \left(\frac{P_{TLT,t}}{P_{HYG,t}} - 1 \right) \times 100$$

- Investment grade spread proxy:

$$\text{IG_spread}_t = \left(\frac{P_{TLT,t}}{P_{LQD,t}} - 1 \right) \times 100$$

- Credit spread level: average of HY and IG spread proxies
- 1-month change: change in credit spread level over the prior 21 trading days

These macro features are later merged with factor return histories to form the input design matrix for factor premia forecasting.

3.3 Factor Construction

MarketMancer focuses on four classic equity styles: **Value**, **Quality**, **Momentum**, and **Low Volatility**. Each factor is constructed from the fundamental and price inputs described above.

3.3.1 Cross Sectional Standardization

For factor construction, raw metrics are standardized cross sectionally within the chosen universe on each date. For a metric $x_{i,t}$ for stock i at time t :

$$z_{i,t}(x) = \frac{x_{i,t} - \bar{x}_t}{\sigma_t(x)}$$

where $\bar{x}_t$ and $\sigma_t(x)$ are the cross sectional mean and standard deviation across all stocks in the universe on date t . This produces comparable z-scores across different metrics and stabilizes the later portfolio construction.

3.3.2 Value Factor

The Value score combines three valuation ratios:

- Book-to-price: $BP_{i,t}$
- Earnings-to-price: $EP_{i,t}$
- Free cash flow-to-price: $FCFP_{i,t}$

Each ratio is standardized cross sectionally on date t , and the composite Value score is...

$$\text{Value}_{i,t} = \frac{1}{3} [z_{i,t}(BP) + z_{i,t}(EP) + z_{i,t}(FCFP)]$$

Higher scores indicate stocks that are cheap relative to the universe on multiple valuation dimensions.

3.3.3 Quality Factor

The **Quality** score aggregates five dimensions of profitability and balance sheet strength:

- Return on equity $ROE_{i,t}$
- Return on assets $ROA_{i,t}$
- Gross margin $GM_{i,t}$
- Free cash flow margin $FCFMargin_{i,t}$
- Inverse leverage $LevInv_{i,t}$

Each is standardized cross sectionally, and the composite Quality score is:

$$\text{Quality}_{i,t} = \frac{1}{5} [z_{i,t}(ROE) + z_{i,t}(ROA) + z_{i,t}(GM) + z_{i,t}(FCFMargin) + z_{i,t}(LevInv)]$$

Higher scores correspond to firms with stronger profitability, better margins, and lower leverage relative to peers.

3.3.4 Momentum Factor

The Momentum factor is built from a short to medium horizon price trend. For each stock, the tool computes the 60 day cumulative log return:

$$\text{MomRaw}_{i,t} = \sum_{k=0}^{59} r_{i,t-k}^{\log}$$

where $r_{i,t-k}^{\log}$ is the daily log return. This raw momentum signal is then standardized across the universe:

$$\text{Momentum}_{i,t} = z_{i,t}(\text{MomRaw})$$

A higher Momentum score means the stock has had stronger recent performance over the past 60 trading days compared to other names in the universe.

3.3.5 Low Volatility Factor

The Low Volatility factor is based on realized volatility over the same 60 day window. For each stock, the tool computes the 60 day realized volatility as...

$$\sigma_{i,t}^{60d} = \text{stdev}(r_{i,t-59}, \dots, r_{i,t})$$

where $r_{i,t}$ are daily returns. To reward lower risk, the model inverts this measure before standardization:

$$\text{LowVol}_{i,t} = z_{i,t}(-\sigma_{i,t}^{60d})$$

Higher LowVol scores therefore indicate stocks with more stable returns and lower realized volatility over the past 60 trading days relative to the rest of the universe.

3.4 Factor Premia Forecasting

With factor return histories in place, MarketMancer estimates how attractive each factor looks over the next H days, both on its own and conditional on macro conditions.

3.4.1 Feature Design

For each factor f and date t , the forecasting models use two groups of predictors:

- A factor history vector $H_f(t)$ that captures persistence and local trends:
 - Recent lagged factor premia $y_{f,t-1}, y_{f,t-5}, y_{f,t-21}, y_{f,t-63}$
 - Moving averages of factor premia over windows such as 21 and 63 days
- A macro state vector $M(t)$ built from the data in Section 3.2.4:
 - Rates level and 1-month change
 - Term spreads (10y-2y) and (10y-3m)
 - VIX level, moving average, and percentile
 - Credit spread level and 1-month change

These predictors are combined into a single feature vector $X_{f,t} = [H_f(t), M(t)]$ that is used by the macro-aware models. For the autoregressive benchmark, only the factor's own history is used.

3.4.2 AR(1) Baseline Model

The primary forecast for each factor uses a simple autoregressive specification in the smoothed H -day premia:

$$y_{f,t} = a_f + b_f y_{f,t-1} + \varepsilon_{f,t}$$

This AR(1) model embodies the idea that factor premia exhibit mechanical persistence: when a factor has recently delivered high (or low) H -day returns, it is more likely to continue in that direction in the near term. The AR(1) model has very few parameters, which keeps it robust to overfitting and easy to interpret.

3.4.3 Macro-Aware Ensemble

In parallel, the system trains a macro-aware ensemble that uses the full feature vector $X_{f,t}$. The ensemble consists of three models:

- **Ridge regression**, a linear model that estimates a weighted combination of the input features. Ridge applies an ℓ_2 penalty that gently shrinks all coefficients toward zero. This keeps the model stable when features are correlated and prevents the weights from becoming excessively large.
- **Lasso regression**, another linear model but with an ℓ_1 penalty. Unlike Ridge, Lasso can shrink some coefficients all the way to zero, meaning it automatically selects a small subset of features that matter most. This produces sparse, interpretable models that often generalize better out of sample.
- **Random Forest**, a non-linear model made up of many decision trees. Each tree sequentially splits the data based on thresholds of macro or lagged-return variables (for example, “is VIX above X?”). The forest averages hundreds of these trees, allowing it to capture interactions, non-linearities, and regime-dependent behavior that linear models cannot express.

Each model produces a forecast $\hat{y}_{f,t}^{(m)}$. The ensemble prediction is the simple average:

$$\hat{y}^{ens} * f, t = \frac{1}{3} (\hat{y}^{ridge} * f, t + \hat{y}^{lasso} * f, t + \hat{y}^{rf} * f, t)$$

In practice, the AR(1) model consistently achieves higher hit rates and lower errors in walk-forward tests, so it is used as the main factor premia input for portfolio construction. The ensemble is most useful as a way to understand how macro conditions relate to factor performance.

3.4.4 Interpreting Macro Drivers

The Random Forest component provides feature importance measures that highlight which variables in $X_{f,t}$ have historically been most predictive of each factor. Specifically, it provides these feature importance scores based on the average reduction in node impurity that each variable in $X_{f,t}$ delivers across all trees. In

practical terms, “most predictive” means the features whose splits most improve the model’s fit and reduce forecast error when predicting forward factor premia. The app surfaces these as the “top drivers” for each factor in each regime, typically including:

- Term spreads and credit spreads
- VIX level and percentile
- Longer-term moving averages of the factor itself

These importance scores are interpreted as predictive associations, not causal relationships, but they give users a concrete sense of which macro dimensions matter most for each style.

3.4.5 Walk Forward Validation Procedure

To evaluate out-of-sample performance, both the AR(1) model and the macro aware ensemble are tested using a fixed 5-fold walk forward (expanding window) procedure.

Starting from the factor premia history and feature matrix:

1. The data is ordered by date and any rows with missing targets are dropped.

Let N be the number of remaining observations and let $n_{\text{splits}} = 5$.

The length of each test block is computed as

$$\text{test_size} = \left\lfloor \frac{N}{n_{\text{splits}} + 1} \right\rfloor$$

2. The sample is then split into six contiguous time blocks of length `test_size` (plus a small remainder, if any):

- The first block is used only for initial training.
- The next five blocks serve as non-overlapping test windows, one for each fold.

For fold $i = 0, 1, 2, 3, 4$:

- The training window uses all observations from the start of the sample up to the end of block $i + 1$.

This creates an expanding window: each new fold adds a full block of history to the training set and never drops earlier data.

- The test window is the next block (block $i + 2$), which has not been used in training in any way.
- On each fold:
 - The ensemble models (Ridge, Lasso, Random Forest) are re-fit on the current training window and used to forecast the forward (H)-day factor premia in the test window.
 - The AR(1) model is also re-estimated using only lagged premia from the same training window and then applied to the test window.

By construction, training and test periods never overlap, and every forecast is made using only information that would have been available at the time, so all evaluation is strictly out of sample.

For each factor and model, the tool then reports:

- Hit rate: fraction of test observations where the predicted sign matches the realized sign of $y_{f,t}$
- Root Mean Squared Error (RMSE)
- Mean Absolute Error (MAE)

These walk forward diagnostics are what ultimately justify using AR(1) as the primary factor premia signal in the results section, since it consistently delivers higher hit rates and lower errors than the macro-aware ensemble across regimes.

3.5 Stock-Level Alpha Modeling

Factor premia forecasts answer the question “Which styles look attractive next?” The alpha model answers “Given those conditions, which individual stocks look attractive within the chosen universe?”

3.5.1 Target: H -day Forward Average Log Return

For each stock i and date t , the alpha model’s target is the H -day forward average log return:

$$\alpha_{i,t} = \frac{1}{H} \sum_{k=1}^H r_{i,t+k}^{\log}$$

This is an absolute return forecast rather than a factor neutral one. The model does not explicitly subtract factor exposures from returns. Instead, it uses factor-related characteristics and technical indicators as features, which allows it to learn return patterns that are indirectly related to factor structure.

3.5.2 Feature Set

The feature vector $X_{i,t}$ for stock i at time t includes:

- **Fundamental features** (mirroring the factor inputs):
 - Book-to-price, earnings-to-price, free cash flow-to-price
 - ROE, ROA
 - Gross margin, free cash flow margin
 - Debt-to-equity and inverse leverage
- **Technical features:**

- 5, 21, and 63 day cumulative log returns (short, medium, and medium long momentum)
- 5, 21, and 63 day rolling volatility (standard deviation of daily returns)

On each modeling date, all features are standardized cross sectionally so that each column has mean zero and unit variance. This standardization makes coefficients comparable and ensures that the regularization treats features on the same scale.

3.5.3 Lasso Specification

The alpha model is a pooled Lasso regression estimated on a panel of stocks and dates. For a given experiment (regime and horizon H), the model solves:

$$\min_{\beta} \sum_{i,t} (\alpha_{i,t} - X'_{i,t}\beta)^2 + \lambda \|\beta\|_1$$

where:

- The sum is over all stocks and dates in the training window
- β is the common coefficient vector
- λ is a regularization parameter chosen by cross validation

Lasso is attractive here because it automatically shrinks many coefficients exactly to zero, leaving a sparse set of active predictors that drive the alpha signal. This keeps the model interpretable and reduces the risk of overfitting to noise.

3.5.4 Training Procedure and Today's Prediction

For each testing regime, the training procedure is:

1. Build a panel dataset by stacking all available stocks and dates over a rolling lookback window (for example about two years) that ends before the evaluation date.

2. Construct the forward H -day log return target $\alpha_{i,t}$ and corresponding feature vectors $X_{i,t}$.
3. Standardize features across the panel and fit a single Lasso model with cross validated λ .

At the most recent date T , the system:

1. Recomputes the current feature vectors $X_{i,T}$ for all stocks in the universe.
2. Applies the same standardization transformation used in training.
3. Uses the stored Lasso model to generate expected H -day alphas $\hat{\alpha}_{i,T}$.

These predictions form the primary stock-level alpha outputs shown in the app.

3.5.5 Fundamental Score and Top Drivers

To give users more intuition, the tool computes an additional fundamental score for each stock:

$$\text{FundamentalScore}_{i,T} = \frac{1}{2}(\text{ValueZ}_{i,T} + \text{QualityZ}_{i,T})$$

where ValueZ and QualityZ are the standardized Value and Quality scores for stock i at date T . This is displayed alongside the predicted alpha so the user can see whether the signal aligns with strong fundamentals or is driven mainly by technicals.

For the top predicted alpha names, the app also lists the top feature drivers: the standardized features with the largest nonzero Lasso coefficients in absolute value.

Since all features are standardized, a positive coefficient $\beta_j > 0$ means:

- A one standard deviation increase in feature j is associated with an increase of β_j in expected H -day return.

Similarly, a negative coefficient $\beta_j < 0$ means that high values of this feature are associated with lower expected H -day returns. This provides a direct explanation of why the model is bullish or bearish on each highlighted stock.

Finally, the tool computes a simple cross sectional alpha strength measure: the difference between the average expected alpha of the top decile of stocks and the bottom decile. This spread summarizes how much separation the model is creating between attractive and unattractive names.

3.6 Unified Portfolio Optimization

The last step is to turn the alpha predictions into a single, coherent long short portfolio that reflects the model’s views while remaining stable and interpretable.

3.6.1 Expected Alpha Vector

At the most recent date T , the alpha model produces an expected H -day return $\hat{\alpha}_{i,T}$ for each stock in the universe. Collecting these into a vector:

$$\mu_i = \hat{\alpha}_{i,T}$$

gives the expected return input for the portfolio stage.

3.6.2 Covariance Estimation and Risk View

To get a realistic view of portfolio risk, the system estimates a Ledoit–Wolf shrunk covariance matrix of recent stock returns. Using roughly the last 252 trading days, it first forms the sample covariance matrix S and then blends it with a structured target matrix F :

$$\hat{\Sigma} = (1 - \delta)S + \delta F$$

Here $\delta \in [0, 1]$ is a shrinkage intensity chosen analytically rather than by hand.

Following Ledoit and Wolf’s “Honey, I Shrunk the Sample Covariance Matrix” (2004), the procedure selects δ to minimize the expected squared distance between $\hat{\Sigma}$ and the unknown true covariance matrix in Frobenius norm. Intuitively, when the

sample is noisy or the cross-section is large relative to the history, the method leans more toward the stable target F ; when the data are rich, it relies more on the empirical covariance S .

This matters because raw sample covariances are notoriously unstable in exactly the setting MarketMancer operates in: tens to hundreds of stocks, limited lookback, and time-varying correlations. Ledoit–Wolf shrinkage produces a well-conditioned, positive-definite risk matrix that is far less sensitive to estimation error, which in turn makes portfolio-level volatilities and correlations more reliable.

In the current version of the tool, this shrunk covariance matrix is used primarily to interpret the volatility, concentration, and diversification of the unified portfolio rather than to run a full mean-variance optimizer. It provides an institutionally credible risk lens while keeping the actual weighting rule simple and transparent.

3.6.3 Long Short Weighting Rules and Constraints

Given the expected alpha vector μ , MarketMancer applies a deliberately simple weighting rule that is easy to explain:

- **Direction:**
 - Go long stocks with $\mu_i > 0$
 - Go short stocks with $\mu_i < 0$
- **Scaling:**
 - Scale longs and shorts so that:
 - No individual stock exceeds a 10 percent absolute weight:

$$|w_i| \leq 0.10$$

- Total **gross exposure** (sum of absolute weights) is capped at 1.5x:

$$\sum_i |w_i| \leq 1.5$$

- Within these constraints, the portfolio is approximately dollar neutral, with long and short exposures of similar magnitude.

This rule keeps the weights aligned with the relative strength of the alpha signals but avoids the instability that can come from aggressive quadratic optimization on noisy inputs. Expected portfolio H -day alpha is computed as:

$$\hat{\alpha}_T^{port} = \sum_i w_i, \hat{\alpha}_{i,T}$$

and portfolio risk measures are derived from $\hat{\Sigma}$. The unified portfolio outputs in the app show:

- The tickers, weights, and sides (long or short)
- The expected H -day portfolio alpha
- Diagnostics that link these positions back to factor views and stock-level drivers

Together, these steps complete the pipeline from user inputs and raw data to an integrated set of factor views, alpha signals, and a unified long short portfolio

4. Results

4.1 Testing Regimes

We evaluate MarketMancer across three regimes, each defined by its universe size, sample window, and forecast horizon. This allows us to see how stability, predictability, and portfolio outputs change as more data becomes available. Table 1 summarizes the three testing regimes, showing the universe size, sample window, and chosen H -day forecast horizon for each.

Table 1. Testing Regime Definitions

Regime	Universe Size	Sample Window	H -day Horizon
Small	10 stocks	Jan 2025 onward	10 days
Medium	50 stocks	Jan 2022 onward	21 days
Large	250 stocks	Jan 2019 onward	42 days

All tests use seed = 42 for reproducibility.

4.2 Historical Factor Behavior Across Regimes

Table 2 reports annualized returns, volatilities, and Sharpe ratios for each factor in the Small, Medium, and Large regimes to compare how factor behavior changes with sample length and universe size. Across the three regimes, factor performance shows intuitive and economically plausible patterns.

Table 2. Historical Factor Behavior Across Regimes

Factor	Ann. Return % (S)	Ann. Vol % (S)	Sharpe (S)	Ann. Return % (M)	Ann. Vol % (M)	Sharpe (M)	Ann. Return % (L)	Ann. Vol % (L)	Sharpe (L)
Low Vol	-3.58	17.24	-0.13	-9.28	13.62	-0.65	-5.46	11.04	-0.45
Momentum	17.47	20.10	0.90	11.95	13.49	0.90	4.85	6.37	0.78
Quality	-5.03	15.48	-0.26	-1.74	11.82	-0.09	3.47	8.42	0.45
Value	2.91	19.16	0.24	-15.87	11.17	-1.49	-9.65	8.37	-1.17

Key Observations

- Momentum is consistently the strongest factor across regimes.
- Value turns sharply negative as earlier years enter the sample.
- Low Volatility is negative across all regimes.
- Volatility declines monotonically as the universe size increases.

These patterns validate that the constructed factor portfolios behave as expected before being used for forecasting.

4.3 Factor Premia Forecasts Across Regimes

Table 3 compares the expected H -day factor premia from the AR(1) model and the macro-aware ensemble across all factors and regimes.

Table 3. Factor Premia Forecasts Across Regimes

Factor	AR(1) Exp. Premium % (S)	Ensemble Exp. Premium % (S)	AR(1) Exp. Premium % (M)	Ensemble Exp. Premium % (M)	AR(1) Exp. Premium % (L)	Ensemble Exp. Premium % (L)
Low Vol	-0.34	-0.22	-0.02	-0.04	-0.07	0.00
Momentum	0.59	0.44	0.18	0.13	0.50	0.21
Quality	-0.10	-0.02	-0.04	-0.02	-0.13	-0.02
Value	0.24	0.19	0.03	-0.12	-0.06	-0.07

Key Observations

- Momentum remains positive across all regimes.
- Value weakens materially in the Medium and Large regimes.
- Low Volatility and Quality remain neutral or negative in all windows.
- AR(1) remains the more stable and persistent forecaster.

4.4 Top Drivers per Factor

Table 4 lists the top macro and factor-history drivers for each factor in each regime, based on Random Forest feature importance.

Table 4. Top Drivers per Factor Across Regimes

Factor	Small Regime Top Drivers	Medium Regime Top Drivers	Large Regime Top Drivers
Low Vol	vix_close, vix_percentile, LOW_VOL_ma21	LOW_VOL_ma63, term_spread_10y2y, credit_spread_level	rates_level, LOW_VOL_ma63, credit_spread_level
Momentum	MOMENTUM_ma21, term_spread_10y3m, MOMENTUM_ma63	term_spread_10y2y, credit_spread_level, term_spread_10y3m	credit_spread_level, rates_level, term_spread_10y2y
Quality	vix_percentile, QUALITY_ma63, credit_spread_level	term_spread_10y2y, QUALITY_ma63, term_spread_10y3m	term_spread_10y2y, QUALITY_ma63, rates_1m_change
Value	VALUE_ma63, credit_spread_1m_change, vix_close	term_spread_10y2y, VALUE_ma63, credit_spread_level	rates_level, credit_spread_level, term_spread_10y3m

Key Observations

- Small regime relies more on recent volatility and factor own-history (VIX and short-horizon moving averages)
- Medium and large regimes shift toward broader macro conditions, with rates and spreads becoming dominant drivers
- These drivers align well with known economic interpretations.

4.5 Walk-Forward Validation Across Regimes

Table 5 shows walk-forward performance for AR(1) and the ensemble, including hit rates and RMSE/MAE for every factor and regime.

Table 5. Walk-Forward Validation Across Regimes

Factor	AR(1) Hit / RMSE / MAE	Ensemble Hit / RMSE / MAE	AR(1) Hit / RMSE / MAE	Ensemble Hit / RMSE / MAE	AR(1) Hit / RMSE / MAE	Ensemble Hit / RMSE / MAE
	(S)	(S)	(M)	(M)	(L)	(L)
Low Vol	81.73% / 0.0014 / 0.0012	31.73% / 0.0042 / 0.0039	93.87% / 0.0006 / 0.0004	50.67% / 0.0023 / 0.0019	94.04% / 0.0002 / 0.0002	55.51% / 0.0013 / 0.0011
Momentum	82.69% / 0.0016 / 0.0013	71.15% / 0.0030 / 0.0024	89.47% / 0.0006 / 0.0004	50.13% / 0.0021 / 0.0018	92.65% / 0.0001 / 0.0001	43.31% / 0.0011 / 0.0008
Quality	87.50% / 0.0012 / 0.0010	57.69% / 0.0032 / 0.0028	90.93% / 0.0005 / 0.0004	51.20% / 0.0017 / 0.0014	94.19% / 0.0002 / 0.0001	52.43% / 0.0011 / 0.0010
Value	84.62% / 0.0016 / 0.0013	42.31% / 0.0039 / 0.0035	94.13% / 0.0005 / 0.0004	62.27% / 0.0018 / 0.0015	94.63% / 0.0002 / 0.0001	65.07% / 0.0013 / 0.0011

Key Observations

- AR(1) significantly outperforms ensemble in hit rate across all factors and regimes.
- Forecast errors (RMSE, MAE) drop sharply in larger regimes, likely due to their richer history.
- Ensemble improves with more data but remains less stable than AR(1).
- Overall, AR(1) is the more reliable factor forecaster.

4.6 Unified Portfolio Alpha Across Regimes

Table 6 reports the expected H -day alpha of the unified portfolio in each regime, along with the corresponding average one-day return.

Table 6. Unified Portfolio Alpha Across Regimes

Regime	H-day Horizon	Expected H-day Alpha (%)	1-day Equivalent (% per day)
Small	10 days	1.28	0.128
Medium	21 days	2.02	0.096
Large	42 days	7.93	0.189

Key Observations

- Large regime produces the strongest signals, both daily and cumulative.
- Medium regime has the weakest daily alpha despite better H -day returns than Small.
- Small regime sits between Medium and Large on a daily basis despite having the lowest overall H -day alpha.

Overall, these results reflect the increased reliability and richness of signals as the universe and history expand.

5. Discussion and Conclusions

MarketMancer set out to answer a simple but surprisingly difficult question: can we connect macro conditions, factor signals, and stock-level characteristics in a single framework that produces forecasts and a portfolio users can actually understand? The results across all three regimes suggest that the answer is yes, at least within the boundaries of the data and models used here.

Across the Small, Medium, and Large regimes, the factor portfolios behaved in ways that were intuitive and economically sensible. Momentum was reliably strong, Low Volatility was defensive but negative on average, Quality improved as the sample widened, and Value weakened materially once earlier years entered the window. This gave us confidence that the factor construction layer was working as intended before any forecasting was introduced.

On the forecasting side, the AR(1) model consistently outperformed the macro-aware ensemble on hit rate, sometimes by a very wide margin. This was true for every factor and every regime. The ensemble was still informative because it highlighted which macro variables historically mattered most, but its forecasts were noisier and far less stable than the simple AR(1). This contrast reinforced a theme that showed up repeatedly throughout the project: when the sample size is small relative to the number of predictors and correlations shift over time, simpler models often win.

The top driver results also told a coherent story. In the Small regime, where the data window is short, recent volatility, VIX measures, and short horizon factor lags dominated. In the Medium and Large regimes, broader macro variables such as the term spread, rates level, and credit spreads moved to the top of the rankings. This progression made sense and matched our intuition about how the predictive structure should deepen as the sample grows.

Stock-level alpha predictions showed meaningful cross sectional dispersion across all regimes, and the unified long short portfolio translated these signals into clear, interpretable positions. The expected H -day portfolio alpha increased dramatically in the Large regime, reflecting the benefits of a longer history and a richer factor environment. Even in the smaller regimes, the portfolio construction step produced diversified allocations that were easy to explain and tied directly to the strength of each stock's predicted alpha.

Taken together, these results demonstrate that MarketMancer’s layered pipeline works as intended. Each component adds something distinct, and the end product behaves like a coherent investment process rather than a collection of disconnected modules. The system does not solve every problem in factor investing, but it succeeds in providing a transparent, unified framework that links macro regimes, factor dynamics, stock characteristics, and portfolio outputs in a way that users can follow from start to finish.

6. Suggestions for Future Research

The project opens several avenues for further development, both methodologically and in terms of practical enhancements.

First, richer macro models could be explored. The ensemble here was intentionally simple, but tree based methods, interaction terms, or regime switching models might capture non linear structure without overfitting if applied carefully. Another natural extension is to allow the AR(1) baseline to incorporate volatility adjusted innovations or rolling structural break detection.

Second, the alpha model could be expanded. Lasso works well for interpretability, but alternative estimators such as elastic net, gradient boosted trees, or even shallow neural networks might extract additional signal, especially if combined with stronger regularization and walk forward controls. Feature engineering using PCA or sector neutral residualization may also improve out of sample stability.

Third, the portfolio stage could be made more sophisticated. The current risk adjusted scoring rule is intentionally simple, but users could benefit from optional enhancements such as:

- a full mean variance optimizer using the Ledoit Wolf covariance matrix

- turnover controls
- transaction cost modeling
- factor exposure constraints or hedging overlays.

Each of these additions would make the system feel more like an institutional quality portfolio engine while preserving the glass box philosophy.

Fourth, additional data sources could reduce bias. The tool currently depends on the FMP API and uses current S&P 500 membership, which introduces survivorship bias. Expanding to full historical index membership, adding Compustat point in time data, or integrating alternative datasets (options implied volatility, earnings call sentiment, ETF flows) would meaningfully broaden the modeling landscape.

Finally, user experience could be deepened. Interactive scenario tests, explanations that trace how each model layer influenced the final portfolio, and a sandbox mode that allows users to toggle model assumptions on and off would all help the tool function as a teaching device as well as a forecasting engine.

Overall, MarketMancer provides a strong foundation. Each layer produced coherent and interpretable results, and the integrated pipeline proved both functional and informative. The next steps involve building on this foundation with better data, richer models, and a more flexible portfolio layer, while continuing to keep transparency and interpretability at the center of the design.

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Appendices

Appendix A: Walk-Forward Validation Diagram

This appendix illustrates the five-fold expanding-window walk-forward procedure used in all factor premia forecasting. The diagram below shows a simple example with $N = 600$ observations.

To determine the length of each test block, we use:

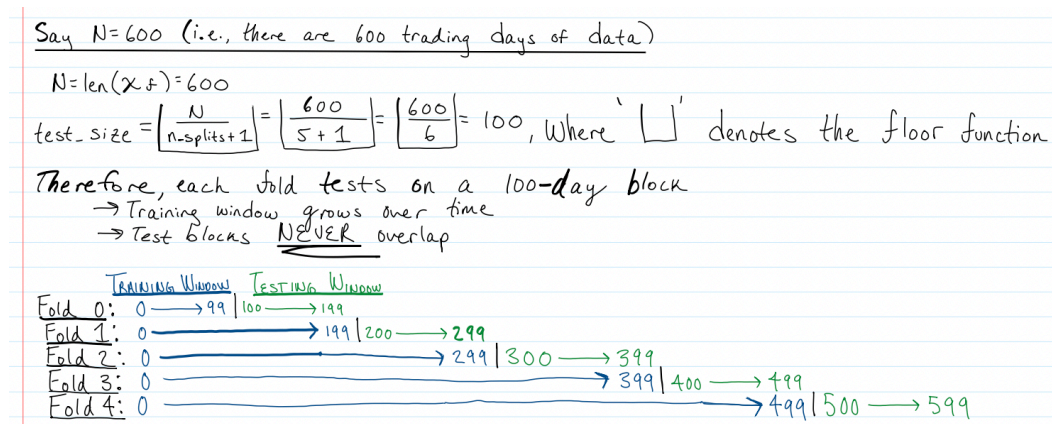
$$\text{test_size} = \left\lfloor \frac{N}{n_{\text{splits}} + 1} \right\rfloor = \left\lfloor \frac{600}{6} \right\rfloor = 100.$$

The sample is therefore divided into six consecutive 100-day blocks.

Fold i trains on all data up through block $i + 1$ and tests on block $i + 2$.

The training window expands each fold, and the test windows never overlap.

Figure A1. Illustration of 5-Fold Walk-Forward Split with $N = 600$



Appendix B. Illustrative Screenshots of the MarketMancer Interface

This appendix provides visual examples of the MarketMancer interface. Each screenshot corresponds to one stage of the pipeline described in the main text. The images are included for reference only and reflect a single sample run of the tool; results vary with the user's inputs.

B.1 User Inputs and Universe Construction

Description: The sidebar where the user configures the run. At the top, the user enters the FMP API key and selects the start date. Under Universe, the user chooses the universe size, decides whether to randomize the universe selection, and sets the random seed. Under Forecasting, the user selects the H -day forecast horizon and how many top drivers (3 or 5) to display for each factor.

The S&P 500 Sector Benchmark section shows the status of the cached S&P 500 dataset, including the last refresh dates for fundamentals and prices and the number of S&P names currently cached. The checkbox “Use S&P 500 sector-relative scoring”* controls whether stock factor scores are evaluated relative to their S&P 500 sector peers instead of only the selected universe. The Refresh cache buttons (All, Prices, Clear) let the user force an update of the benchmark data or clear the cache. Finally, the Run pipeline button at the bottom triggers the full MarketMancer pipeline with the chosen settings.

Configuration

FMP API Key



.....



Start date (YYYY-MM-DD)



2022-01-01

Universe

Universe size

50



☒ Randomize universe selection

Random seed

42



Forecasting

Forecast horizon (days)



21



Number of top drivers to show



3

5

S&P 500 Sector Benchmark

✓

Fundamentals: 2025-12-11 (monthly)

✓

Prices: 2025-12-11 (daily)

508 S&P 500 stocks cached

✓

Use S&P 500 sector-relative scoring

?

Refresh cache:

All

Prices

Clear

Run pipeline

B.2 Historical Factor Performance

Description:

This panel shows how the long-short factor portfolios have behaved over the full sample. At the top, a “Details” box explains the methodology and the statistics being reported. The summary table reports, for each factor, the total return, annualized return, annualized volatility, and a Sharpe ratio computed relative to the available

daily risk-free series. Below the table, a log-scale chart plots the cumulative growth of one dollar invested in each long-short factor portfolio over time, allowing for direct visual comparison across Value, Quality, Momentum, and Low Volatility.

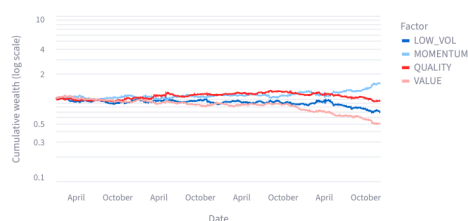
Historical factor performance

How have these factor portfolios actually behaved over time?

Details					
This section summarizes how each long short factor portfolio has performed over the full sample used in the model. For each factor, we show total and annualized returns, annualized volatility, a Sharpe ratio computed relative to the available daily risk free series when present (otherwise zero), and the worst peak to trough drawdown over the period.					

	Factor	Total return %	Annualized return %	Annualized volatility %	Sharpe (rf adj)
0	LOW_VOL	-31.67	-9.28	13.62	-0.65
1	MOMENTUM	55.46	11.95	13.49	0.90
2	QUALITY	-6.64	-1.74	11.82	-0.09
3	VALUE	-49.10	-15.87	11.17	-1.49

Cumulative growth of one unit invested in each factor long short portfolio over time.



B.3 Factor Premia Forecasts

Description:

This tab shows which factors the model expects to outperform over the next H days.

It includes three pieces:

1. **AR(1) Expected Premia** The main table displays each factor's H -day expected return based on a simple AR(1) model. These are the primary forecasts used in the unified portfolio.
2. **Ensemble Forecasts (Expandable Panel)** An optional view comparing the AR(1) outputs with the macro-aware ensemble (Ridge, Lasso, Random Forest). This section is for diagnostics only.
3. **Top Drivers per Factor (Expandable Panel)** A Random Forest importance table showing which macro features and lagged factor signals were most predictive for each factor.

Overall, this tab summarizes the model's current view of factor conditions and the macro forces influencing them.

Factor premia forecasts

Which factor styles look poised to outperform next?

Details		
These forecasts estimate each factor's expected return (in percent) over the next H days, where H is the forecast horizon set by the user. The primary forecast uses a simple AR(1) model on each factor's own history, while macro features are used for diagnostics and driver analysis. Higher values indicate a stronger expected tailwind for that factor (candidates to overweight), while negative values indicate expected headwinds (candidates to underweight).		
Factor	Expected Premium % (AR(1))	
2 MOMENTUM		0.18
0 VALUE		0.03
3 LOW_VOL		-0.02
1 QUALITY		-0.04

Show machine learning ensemble factor forecasts (Ridge, Lasso, Random Forest)			
Factor	Ensemble Premium %	AR(1) Premium %	
2 MOMENTUM	0.13		0.18
1 QUALITY	-0.02		-0.04
3 LOW_VOL	-0.04		-0.02
0 VALUE	-0.12		0.03

Top drivers per factor

What macro forces are actually moving these factors?

Details	
The table below shows which macro features the model found most predictive when forecasting each factor's expected return over the next H days. These macro features include rate levels, term spreads, credit spreads, and volatility measures. More technically, "most predictive" means these features produced the largest reductions in forecast error in the random forest model, with RF importance measuring the average improvement in fit when that feature is used across the trees.	

	Factor	Driver	RF Importance
2	VALUE	credit_spread_level	0.140
3	QUALITY	term_spread_10y2y	0.302
4	QUALITY	QUALITY_ma63	0.225
5	QUALITY	term_spread_10y3m	0.122
6	MOMENTUM	term_spread_10y2y	0.415
7	MOMENTUM	credit_spread_level	0.134
8	MOMENTUM	term_spread_10y3m	0.131
9	LOW_VOL	LOW_VOL_ma63	0.299
10	LOW_VOL	term_spread_10y2y	0.299
11	LOW_VOL	credit_spread_level	0.100

B.4 Factor Construction Outputs

Description:

This section summarizes the factor building process after all raw data has been pulled and standardized. It displays:

- **Universe composition**

The exact tickers included in the user's selected universe.

- **Factor portfolio sizes**

The number of stocks contributing to each long–short factor portfolio (Value, Quality, Momentum, Low Vol).

- **Sector-relative S&P 500 benchmark scores**

For every stock in the user's universe, the tool reports sector-relative percentile scores for Value, Quality, Momentum, and Low Volatility.

These percentiles compare each stock only to other S&P 500 names *within the same sector*, making the scores more interpretable.

- **Benchmark cache diagnostics**

A detailed view of the S&P 500 fundamentals and prices caches:

- How many stocks are stored
- Last refresh date
- Refresh frequency (monthly for fundamentals, daily for prices)
- Whether the cache is stale
- Full list of sectors represented

This tab gives the user a complete view of how factor inputs were built, how the S&P 500 benchmark data supports sector-relative scoring, and whether cached data is up to date.

S&P 500 Benchmark Cache Info	
Fundamentals Cache - Exists: True - Last updated: 2025-12-08 - Stocks: 508 - Refresh cycle: 30 days - Stale: False	Prices Cache - Exists: True - Last updated: 2025-12-08 - Stocks: 502 - Refresh cycle: 1 days - Stale: False
Sectors in cache: Basic Materials, Communication Services, Consumer Cyclical, Consumer Defensive, Energy, Financial Services, Healthcare, Industrials, Real Estate, Technology, Utilities	

- **AR(1) baseline accuracy**

Directional hit rate and numerical forecast errors (RMSE, MAE).

- **Machine-learning ensemble accuracy**

Validation metrics for Ridge, Lasso, and Random Forest, shown alongside the AR(1) benchmarks.

The purpose of this view is to show, transparently, how each model would have performed historically under realistic train-then-test conditions, and why the AR(1) model is used as the primary signal in the pipeline.

Factor signal validation (walk-forward)

Can I trust these factor forecasts out of sample?

Details

This section evaluates how well the forecasting models would have performed historically using walk-forward validation. The data is split into several consecutive train-test windows: for each window, the models are trained on past data and then tested only on the period that comes immediately after it, mimicking real-time forecasting. We report accuracy for two models: a simple AR(1) baseline and a three-model machine-learning ensemble (Ridge, Lasso, Random Forest). For each, we show the direction hit rate (how often the model correctly predicted the sign of the factor's forward return), as well as RMSE and MAE to measure numerical forecast error.

AR(1) baseline performance

	Factor	AR(1) Hit Rate	AR(1) RMSE	AR(1) MAE
0	VALUE	94.13%	0.0005	0.0004
1	QUALITY	90.93%	0.0005	0.0004
2	MOMENTUM	89.47%	0.0006	0.0004
3	LOW_VOL	93.87%	0.0006	0.0004

Show machine learning ensemble validation metrics (Ridge, Lasso, Random Forest)

	Factor	Ensemble Hit Rate	Ensemble RMSE	Ensemble MAE	AR(1) Hit Rate	A
0	VALUE	62.27%	0.0018	0.0015	94.13%	
1	QUALITY	51.20%	0.0017	0.0014	90.93%	
2	MOMENTUM	50.13%	0.0021	0.0018	89.47%	
3	LOW_VOL	50.67%	0.0023	0.0019	93.87%	

B.6 Unified Long–Short Portfolio

Description:

This section displays the unified long–short portfolio constructed from the model's stock-level H -day alpha forecasts. The optimizer assigns weights based on expected alpha and a Ledoit–Wolf covariance estimate, while enforcing practical constraints such as per-stock weight caps and a total gross-exposure limit.

The table reports:

- Each selected stock, its portfolio weight, long or short side, and its own expected H -day alpha
- The portfolio's overall expected H -day alpha, shown beneath the table

This tab brings together all earlier signals and shows the final, trade-ready portfolio implied by the model.

Optimized unified portfolio

Given all signals together, what long/short portfolio should I actually hold?

Details

This section shows a unified long/short portfolio built from the model's H -day stock-level alpha forecasts. The optimizer chooses weights that maximize expected H -day portfolio alpha relative to portfolio risk, where risk is measured using a Ledoit Wolf shrinkage estimate of the recent return covariance matrix. Stocks with higher expected alpha and more favorable risk characteristics receive higher positive weights (long positions), while stocks with negative expected alpha receive negative weights (short positions). For stability, the portfolio enforces practical constraints: no individual position may exceed the per-stock weight cap (currently 10 percent), and total gross exposure is limited (currently at 1.5 times the portfolio's capital). The table reports each stock's portfolio weight, side, and its own expected H -day alpha in percent.

	ticker	weight	side	expected_alpha_%
0	BRK-B	0.032	Long	3.19
1	CAT	0.026	Long	3.92
2	NRG	0.025	Long	6.33
3	AXP	0.024	Long	3.72
4	CAH	0.024	Long	3.22
5	COR	0.024	Long	2.72
6	GOOGL	0.024	Long	3.79
7	PARA	0.024	Long	2.99
8	CMA	0.020	Long	3.28
9	DGX	0.020	Long	2.29

Portfolio expected 21-day alpha: 2.02%

B.7 Stock-Level Alpha Predictions

Description:

This view highlights the top 10 stocks by expected H -day alpha from the Lasso alpha model. It shows:

- Each ticker's predicted H -day excess return (in percent)
- A fundamental score that summarizes its Value and Quality signals
- An alpha signal summary (top-decile mean, bottom-decile mean, and their spread)
- For each of the top 10 names, a small table of the most important standardized features and Lasso coefficients explaining why the model is bullish on that stock

Together, this panel translates the factor and macro work into concrete stock picks and gives a transparent explanation for each recommendation.

Alpha predictions (top 10)

Which individual stocks show the strongest forward alpha right now?

Details

This section lists the 10 stocks with the highest expected *H*-day alpha from a Lasso regression that links standardized stock characteristics (valuation, quality, momentum, size, etc.) to their future *H*-day returns. For each name, the expected alpha is the model's forecast of its *H*-day excess return (in percent) based on those historical relationships, and the fundamental score summarizes how attractive its fundamentals look on those same dimensions.

	ticker	expected_alpha_%	fundamental_score
10	NRG	6.33	0
39	CAT	3.92	2
15	GOOGL	3.79	4
48	AXP	3.72	2
25	CMA	3.28	0
42	CAH	3.22	0
5	CTLT	3.21	-4
7	BRK-B	3.19	0
26	PARA	2.99	0
30	COR	2.72	0

Alpha signal summary: Top decile mean 4.21%, Bottom decile -1.74%, Spread 5.95%.

Top drivers for each of the top 10 stocks

Why is the model bullish or bearish on this stock?

Details

For each stock, the model predicts its expected *H*-day alpha using a Lasso regression trained on historical data. All input features are standardized before estimation, so each coefficient measures how a one standard deviation increase in that feature changes the stock's predicted *H*-day alpha, holding other features fixed. A positive coefficient means higher values of the feature are associated with higher predicted alpha, and a negative coefficient means the opposite.

NRG - 6.33%

	feature	Coefficient
0	mom_63d	-0.000000000000000001
1	mom_5d	0
2	vol_5d	0

CAT - 3.92%

	feature	Coefficient
0	vol_63d	0.000000000000000003
1	mom_5d	0
2	vol_5d	0

GOOGL - 3.79%

	feature	Coefficient
0	mom_5d	0
1	vol_5d	0
2	mom_21d	0

AXP - 3.72%

	feature	Coefficient
0	vol_21d	0.000000000000000005
1	mom_5d	0
2	vol_5d	0

CMA - 3.28%

	feature	Coefficient
0	mom_5d	0
1	vol_5d	0
2	mom_21d	0

CAH - 3.22%

	feature	Coefficient
0	mom_5d	0
1	vol_5d	0
2	mom_21d	0

CTLT - 3.21%

	feature	Coefficient
0	mom_63d	-0.0853
1	mom_21d	0.0417
2	vol_21d	-0.0051

BRK-B - 3.19%

	feature	Coefficient
0	vol_63d	-0.0256
1	vol_5d	0.0089
2	mom_63d	-0.0085

PARA - 2.99%

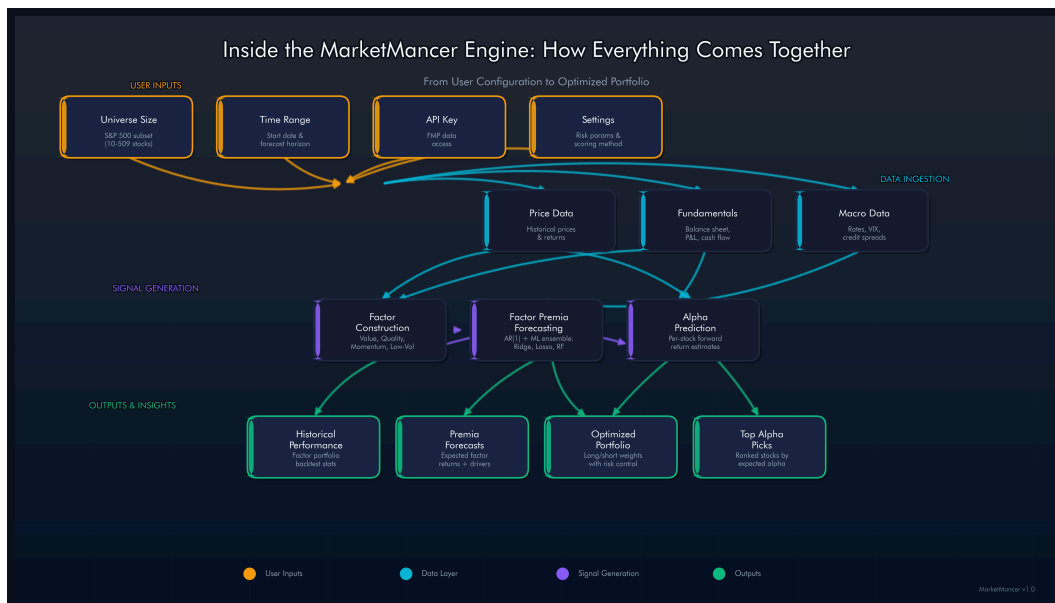
	feature	Coefficient
0	mom_63d	-0.0394
1	vol_63d	-0.0116
2	vol_5d	0

COR - 2.72%

	feature	Coefficient
0	mom_5d	0
1	vol_5d	0
2	mom_21d	0

Appendix C. End-to-End System Flowchart

The diagram below provides a high-level view of how MarketMancer transforms user inputs into factor insights, alpha predictions, and an optimized long short portfolio. It summarizes the full workflow of the engine and shows how each module connects to the next.



C.1 User Inputs (Gold Nodes)

The process begins with the choices the user makes in the sidebar:

- **Universe size** (10–509 stocks)
- **Time range** (start date and forecast horizon)
- **API Key** for accessing data
- **Settings** such as scoring options

These inputs determine what data the system retrieves, which stocks enter the analysis, and the timing assumptions for all forecasts.

C.2 Data Layer (Teal Nodes)

Given the user's configuration, the engine fetches and aligns three categories of data:

- **Price data:** historical prices and returns
- **Fundamentals:** balance sheet, income statement, and cash flow data
- **Macro data:** interest rates, VIX, and credit-spread proxies

These datasets form the backbone for both factor construction and forecasting.

C.3 Signal Generation (Purple Nodes)

The central modeling layer consists of three components:

- **Factor Construction**
 - Builds Value, Quality, Momentum, and Low Volatility signals.
- **Factor Premia Forecasting**
 - Uses AR(1) and a macro-aware ensemble (Ridge, Lasso, Random Forest) to estimate each factor's expected H -day return and its top macro drivers.
- **Alpha Prediction**
 - Uses Lasso regression to estimate expected H -day return for each stock based on standardized fundamentals and technical indicators.

Each block consumes the aligned data and outputs the signals that feed into the final portfolio.

C.4 Outputs and Insights (Green Nodes)

The final layer shows what the user sees after running the pipeline:

- **Historical Performance**
 - Backtested behavior of the four factor portfolios.

- **Premia Forecasts**
 - Expected factor returns and their top macro drivers.
- **Optimized Portfolio**
 - Long and short weights constructed using predicted alphas and Ledoit Wolf covariance shrinkage.
- **Top Alpha Picks**
 - Ranked stocks with the strongest expected H -day returns and their key model drivers.

How to Use This Diagram

This flowchart serves as a quick reference for understanding how MarketMancer’s components interact.

It helps readers:

- visualize the logic behind the system architecture,
- see the separation between data ingestion, signal generation, and outputs,
- and follow the “story” of how raw data becomes an actionable portfolio.

It complements Section 2 (System Overview) and Section 3 (Methodology), giving a visual summary of the same workflow.